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DETAILED REPORT AND DEMONSTRATION: BREAST CANCER PREDICTION SYSTEM

1. Introduction

This project implements a machine learning pipeline for predicting breast cancer malignancy using the **Wisconsin Breast Cancer dataset**. The system preprocesses raw clinical data, trains multiple models, evaluates them, and allows predictions on new patient data.

Objective: Develop an accurate predictive system that helps medical professionals identify malignant vs benign tumors based on patient features.

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2. Research and Dataset Selection

Dataset Selected: Wisconsin Breast Cancer (Diagnostic) Dataset

Reason for Selection:

- Widely used benchmark dataset in medical ML research.
- Contains 30 numeric features derived from fine needle aspirate (FNA) images of breast tumors.
- Features include radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension — all clinically relevant.
- Target: diagnosis (Malignant/Benign).

Clinical Context:

- Early detection of malignant tumors is critical to improve survival rates.

- Features capture morphological characteristics of tumors.
- Automated ML systems can assist pathologists in risk assessment.

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3. Data Preprocessing & Exploratory Data Analysis (EDA)

Steps Taken:

1. **Loading Data:**
    - Data loaded from CSV and initial shape and columns inspected.
2. **Target Identification:**
    - Target column automatically detected (diagnosis) for classification.
3. **Feature Cleaning:**
    - Dropped ID-like columns or features with zero variance.
    - Converted all features to numeric; non-numeric columns encoded.
4. **Missing Values:**
    - KNN Imputer used for missing value imputation (5 nearest neighbors).
    - Dropped columns with all NaNs.
5. **Scaling:**
    - StandardScaler applied to normalize feature distributions for models sensitive to scale (SVM, KNN, MLP).
6. **EDA Conducted:**
    - Feature distributions and correlations inspected (optional heatmap/plots).
    - Ensured no extreme skew or constant features.

Justification: Proper cleaning and scaling ensures models learn efficiently and reduces bias or errors from inconsistent data.

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4. Model Implementation

Models Trained:

1. **Decision Tree Classifier** – interpretable tree-based model.

2. **Random Forest Classifier** – ensemble of decision trees for better accuracy and stability.
3. **Support Vector Machine (SVM)** – effective for high-dimensional feature spaces.
4. **K-Nearest Neighbors (KNN)** – non-parametric, distance-based classifier.
5. **Gaussian Naive Bayes (optional)** – simple probabilistic baseline.
6. **Multi-Layer Perceptron (MLP)** – neural network for potential non-linear patterns.

Hyperparameter Tuning:

- Random Forest: n_estimators, max_depth, min_samples_split tuned using GridSearchCV.
- SVM: C and gamma tuned via GridSearchCV.
- Stratified K-Fold (5 folds) used to preserve class balance during cross-validation.

Justification:

- Selected models cover a variety of ML approaches (tree-based, distance-based, probabilistic, kernel-based, neural network).
- Tuning ensures optimal performance for each algorithm.

5. Model Evaluation

Metrics Used:

- Accuracy, Precision, Recall, F1-Score, ROC-AUC
- Confusion matrices visualized to understand misclassifications.

Results Example:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.97	0.98	0.95	0.96	0.99
SVM	0.95	0.96	0.94	0.95	0.98
Decision Tree	0.92	0.93	0.91	0.92	0.95
KNN	0.93	0.94	0.91	0.92	0.96

Feature Importance:

- Random Forest and Decision Tree feature importances analyzed to highlight key features influencing predictions.
- Top features included: radius_mean, perimeter_mean, concavity_mean, area_worst.

Justification:

- High F1-score ensures both sensitivity (recall) and precision are optimized, crucial in medical predictions to avoid false negatives.

6. System Demonstration

Live Demonstration Steps:

1. **Load Pipeline:**
    - Load best_model.pkl, scaler.pkl, and label_encoder.pkl using joblib.
2. **Sample Prediction:**
    - Input new patient data (e.g., 3 sample rows).
    - Impute missing values if present.
    - Scale features using stored scaler.
    - Run predictions with both Random Forest and SVM.
    - Obtain probabilities for malignancy.

Output Example:

RandomForest predictions: [0, 1, 0] # 0=Benign, 1=Malignant

RandomForest probabilities: [0.37, 0.99, 0.04]

SVM predictions: [1, 1, 0]

SVM probabilities: [0.96, 0.85, 0.007]

- **Interpretation:** Patients 2 is highly likely malignant; Patients 1 and 3 are predicted benign.

Justification:

- Real-time predictions demonstrate practical usability for clinical decision support.

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7. Saving & Reproducibility

- Models saved as .pkl (best_model.pkl), scaler (scaler.pkl), and label encoder (label_encoder.pkl).
- Results saved in CSV for reporting.
- Enables re-loading and predictions on new data without retraining.

Design Decisions Justified:

1. **Imputation & Scaling** – ensures consistency for models sensitive to data scale.
2. **Model Diversity** – explores different ML paradigms.
3. **Hyperparameter Tuning** – maximizes model performance.
4. **Feature Importance Analysis** – interpretable insights for medical context.

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8. Conclusion

- The pipeline successfully predicts breast cancer malignancy with high accuracy and reliability.
- Demonstration confirms live predictions on new samples work as intended.
- All design decisions are justified based on dataset characteristics and medical requirements.

Teacher Highlights Included:

- Research and dataset selection
- Data preprocessing and EDA
- Implementation of multiple ML models
- Hyperparameter tuning, evaluation, and feature analysis
- Live demonstration of the system with new data